NAVNEET KUMAR MISHRA

ELECTRONICS & COMMUNICATION ENGINEERING

LinkedIn profile | GitHub

ACADEMIC PROFILE

Degree/Certificate	Institution	Percentage/CGPA	Year/Expected
B-Tech	Government Engineering College Kishanganj	Pursuing	2028
XIIth (BSEB)	Samastipur College Samastipur	67.4%	2024
Xth (CBSE)	Progressive Central High School Patsa	86.4%	2022

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python
- **Embedded System:** Embedded C, Arduino, ESP32, STM32, Raspberry Pi
- **Communication Protocols:** UART, I^{2}C, SPI, MQTT
- **Robotics & Automation:** ROS2, IoT, Sensor & Actuator Integration, Autonomous Robotics
- **CAD, Simulation & Design:** SOLIDWORKS, MATLAB, 3D Printing, Proteus Design
- **Development Tools:** Git, GitHub, Arduino IDE, STM32CudeIDE, PlatformIO, VS Code
- **Areas of Interest:** Robotics, Embedded Systems, Industrial Automation, Drone System

EXPERIENCE & LEADERSHIP

Student Coordinator: e-Yantra Robotics Lab — Feb 2026 - Present

Government Engineering College Kishanganj

- Coordinated robotics laboratory activity and technical sessions for interested student.
- Guided students in Embedded Systems, Arduino, and Robotics.
- Assisted in conducting hands-on experiments, project development and Workshops.
- Supported participation in robotics competitions and technical events.

Technical Internship

Bharat Sanchar Nigam Limited (BSNL) — Dec 2025 – Jan 2026

- Gained hands-on exposure to telecommunication networks, optical fibers communication, switching, and transmission systems.

PROJECTS

1. IoT Controlled Mobile Robotic Arm *ESP32 | SOLIDWORK | Embedded C | IoT | ESP32-CAM* *Video* — Feb 2026 – May 2026

- Designed and develop a custom robotic arm at the e-Yantra Robotics Lab using **SOLIDWORKS** and **3D printing**.
- Developed an ESP32-powered mobile robotic platform with sensor and actuator integration.
- Developed a responsive web-based dashboard for wireless robot control and **real-time system monitoring**.
- Integrated ESP32-CAM for live **video streaming** and remote visual monitoring.
- **Application:** Suitable for remote material handling, industrial automation, and hazardous environment operations.

2. IoT Based Metal Detection Robot *ESP32 | Embedded C | Sensor | IoT* — Feb 2026 – May 2026

- Developed an ESP32-powered IoT-based **metal detection robot** with wireless monitoring and control.
- Integrated a metal detection sensor, mobile robotic platform, and web interface for **real-time operation**.
- Implemented real-time metal detection alerts and remote monitoring through a wireless dashboard.
- Optimized the embedded system for reliable detection, navigation, and **low-power operation**.
- **Application:** Designed for security screening, industrial inspection, and hazardous object detection.

CERTIFICATIONS

View Certificate

- **NPTEL (Elite)** – Embedded System Design with ARM | IIT Kharagpur (SWAYAM)
- **e-Yantra eLSI** – Basics of Embedded Systems & Robotics (Certificate of Merit, 98/100) | IIT Bombay
- **Industrial Internship** – Bharat Sanchar Nigam Limited (BSNL)
- **Cybersecurity Fundamentals** – Tech Mahindra Foundation | Skill India Digital Hub
- **InnovoQuest 2.0** – Innovation & Entrepreneurship Challenge
- **eLSI Sprint** – Embedded Systems & Robotics Challenge
- **Proteus Design Suite** Workshop (Cohort-I) | GEC Kishanganj | Certificate of Participation, Aug 2026

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